

Project Report:

KiputRevive

Team:

Borneo.java

Track:

Non-Technical

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Date:

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Introduction

Background of Case Study: This project addresses Case Study 2: Indigenous Language Preservation. Southeast Asia, particularly Borneo, is home to rich indigenous cultures. However, minority languages like Kiput—spoken by a specific subset of the Orang Ulu community in northern Sarawak—are at severe risk of digital extinction. Because commercial AI and tech platforms focus on high-resource languages, Kiput youth are left without digital tools to engage with their heritage, accelerating the decline of oral traditions.

Chosen SDG: * SDG 4 (Quality Education): Providing inclusive, culturally relevant digital learning tools for indigenous children.

- **SDG 10 (Reduced Inequalities):** Bridging the digital divide by giving representation to a marginalized, low-resource language in the digital space.

Problem Statement: There is a critical lack of interactive, digitized educational resources for the Kiput language. Current language platforms exclude low-resource Bornean languages, forcing indigenous youth to abandon their mother tongue in digital spaces. Furthermore, the lack of a digital dataset makes it impossible to utilize modern AI tools (like Speech-to-Text) to preserve elders' knowledge.

Objectives:

1. To digitize and preserve authentic Kiput vocabulary and oral folktales using native speaker audio.
2. To create an engaging, bilingual (Kiput and Malay/English) UI/UX designed specifically for children aged 5 to 10.
3. To design a culturally safe, interactive prototype that lays the foundational dataset needed for future Kiput AI models.

Review of Existing Solutions

Current Solutions in the Market: Popular language learning applications like Duolingo or Rosetta Stone dominate the market, alongside generic digital flashcard apps.

Gaps & Limitations: These platforms suffer from a massive localization gap. They exclusively serve high-resource languages (like English, Mandarin, or Spanish). They lack

Bornean cultural context, possess zero authentic indigenous audio, and do not accommodate the specific grammatical nuances of languages like Kiput. *KiputRevive* fixes this by using hyper-local, community-sourced audio paired with culturally relevant visual storytelling.

Methodology

Design Thinking Process & User Research:

- **Empathize & Research:** Our primary user research involved direct consultation with a native Kiput speaker from the Orang Ulu community. This ensured 100% authenticity in pronunciation, vocabulary curation, and cultural safety, avoiding the inaccuracies common in AI-translated indigenous content.
 - **Define:** We defined our target audience as youth aged 5 to 10. Research shows that early intervention is crucial for language retention. We determined that a bilingual bridge (Kiput paired with Malay/English) was necessary since these children are already integrated into the national school system.
 - **Ideate:** We conceptualized three core features: interactive vocabulary quizzes, a multimedia Folktale E-book with native audio, and a conceptual "Cultural Guardian" chatbot interface.
 - **Flow Creation:** The user journey was mapped to be child-friendly. A user opens the app, selects a module (e.g., "Story Time"), views visually engaging 2D artwork, reads the bilingual text, and clicks to hear the authentic Kiput pronunciation. The flow prioritizes auditory learning alongside visual reinforcement.
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Results

Outcome: We successfully designed and developed *KiputRevive*, a high-fidelity interactive prototype. We met our primary objectives by successfully digitizing a core set of Kiput vocabulary and integrating a traditional story flow paired with authentic native audio.

By designing a simulated AI chatbot interface, we demonstrated how the app will function as a virtual language mentor. The structured data we gathered (audio paired with bilingual text) serves as a successful proof-of-concept for how communities can manually build the datasets

required to train future Natural Language Processing (NLP) models for endangered languages.

Prototype Showcase

(Note for Freddie: Insert your screenshots here and use these captions!)

- **[Insert Screenshot 1: The Main Dashboard/Menu]**
 - *Caption: The KiputRevive home screen, designed with intuitive, child-friendly navigation for users aged 5 to 10.*
 - **[Insert Screenshot 2: The Folktale E-Book / Audio feature]**
 - *Caption: The interactive storybook feature displaying bilingual text alongside the crucial "Play Audio" button for authentic native Kiput pronunciation.*
 - **[Insert Screenshot 3: The Quiz / Vocabulary Section]**
 - *Caption: Gamified vocabulary quizzes that reinforce language retention through visual and textual cues.*
 - **[Insert Screenshot 4: The AI Chatbot UI]**
 - *Caption: The UI design for our "Cultural Guardian" chatbot, mapping the future integration of a RAG-based AI mentor.*
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AI Acknowledgement

Disclosure of AI Tools Used:

- **Figma:** Utilized for the initial UI/UX wireframing and high-fidelity prototyping of the *KiputRevive* interface. This allowed for rapid iteration of the user journey specifically designed for the 5-10 age demographic.
- **Google Gemini:** Used as a brainstorming partner to structure the design thinking methodology, refine the project report phrasing, and conceptualize prompt structures for future Generative AI integrations.

- **Google Gemini:** Used to generate the 2D storybook illustrations for the UI prototype, bypassing the need for a dedicated graphic artist during the hackathon.

Important Note on Data Authenticity: No AI was used to generate the Kiput language text or audio. To ensure absolute cultural preservation and accuracy, all Kiput data in this prototype was manually transcribed and recorded by a human native speaker.

Conclusion

Summary: *KiputRevive* bridges the critical gap between ancient Bornean heritage and modern digital learning. By prioritizing authentic human data over flawed commercial AI translation, we have created a scalable, engaging prototype that empowers Kiput youth to reclaim their mother tongue.

Future Improvements: Moving forward, we plan to implement a live backend API for the "Cultural Guardian" chatbot using Retrieval-Augmented Generation (RAG) based strictly on our vetted cultural documents. Furthermore, as the app gathers more community-contributed stories, this data will be used to train the first-ever dedicated Speech-to-Text (STT) machine learning model for the Kiput language, eventually scaling this framework to other endangered languages across ASEAN.